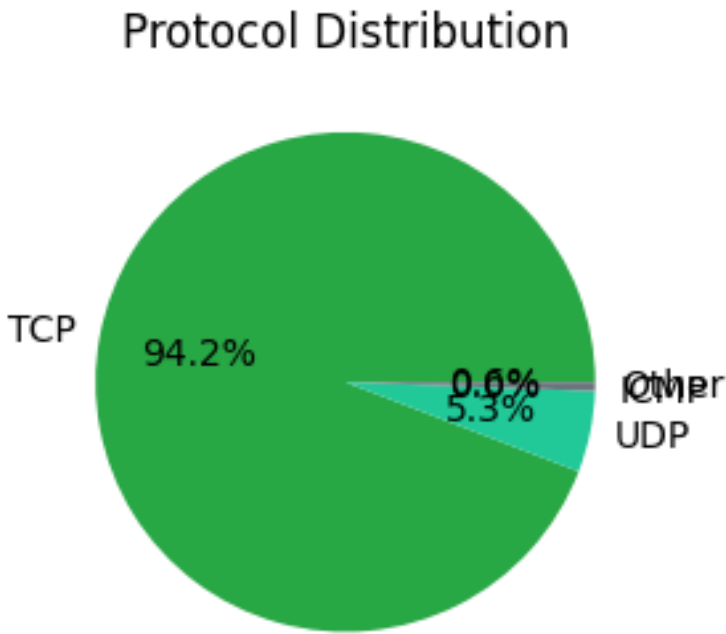


NIDS Session Report

AI-Based Network Intrusion Detection System

Protocol Distribution



Session Summary

Start Time:	1750244178.8857145
Stop Time:	2025-06-18T16:27:19.124501
Uptime:	60 seconds
Packets Processed:	1065
Anomalies Detected:	11
Packet Rate:	0 packets/sec

Protocol Statistics

TCP PACKETS:	1003
UDP PACKETS:	56
ICMP PACKETS:	0
OTHER PACKETS:	6

Alerts (3)

Type: ML_ANOMALY | Time: 2025-06-18T16:26:23.936818

Details: {'anomaly_score': -7.242226722363004, 'features': {'packet_count': 16, 'total_bytes': 989, 'avg_packet_length': np.float64(61.8125), 'std_packet_length': np.float64(4.798681459526148), 'min_packet_length': 60, 'max_packet_length': 75, 'avg_payload_size': np.float64(1.75), 'std_payload_size': np.float64(0.6614378277661477), 'total_payload_size': 28, 'tcp_count': 14, 'udp_count': 0, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 1, 'tcp_ratio': 1.0, 'udp_ratio': 0.0, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 1, 'unique_dst_ports': 14, 'total_ports_accessed': 15, 'common_port_ratio': 0.0, 'high_port_ratio': 1.0, 'avg_dst_port': np.float64(51954.42857142857), 'std_dst_port': np.float64(22.057438932135753), 'min_dst_port': 51930, 'max_dst_port': 51983, 'unique_connections': 6, 'total_connection_attempts': 14, 'avg_packets_per_connection': 2.3333333333333335, 'connection_diversity': 0.375, 'max_packets_per_connection': 3, 'unique_src_ips': 6, 'unique_dst_ips': 1, 'src_ip_diversity': 0.375, 'dst_ip_diversity': 0.0625, 'private_src_ip_ratio': 0.0, 'private_dst_ip_ratio': 1.0, 'packet_rate': 4.499214350531002, 'inter_arrival_time_mean': np.float64(0.23707842826843262), 'inter_arrival_time_std': np.float64(0.30752320338693345), 'burst_count': 8, 'burst_ratio': 0.5333333333333333, 'payload_size_mean': np.float64(1.75), 'payload_size_std': np.float64(0.6614378277661477), 'zero_payload_ratio': 0.125, 'large_payload_ratio': 0.0, 'payload_size_entropy': np.float64(0.5435644431995964), 'syn_count': 14, 'fin_count': 0, 'rst_count': 0, 'psh_count': 0, 'ack_count': 14, 'urg_count': 0, 'syn_ratio': 1.0, 'fin_ratio': 0.0, 'rst_ratio': 0.0, 'psh_ratio': 0.0, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:26:23.936802', 'packet_count': 16, 'protocol_distribution': {'tcp': 14, 'udp': 0, 'icmp': 0}, 'connection_info': {'unique_connections': 6, 'unique_src_ips': 6, 'unique_dst_ips': 1}}

Explanation: The machine learning model detected traffic that is statistically unusual compared to normal patterns.

Type: PORT_SCAN | Time: 2025-06-18T16:26:26.351541

Details: {'src_ip': '10.0.2.15', 'dst_ip': '8.8.4.4', 'dst_port': 443, 'protocol': 17}

Explanation: A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.

Type: DDOS_INDICATOR | Time: 2025-06-18T16:26:59.078604

Details: {'src_ip': '75.2.76.8', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Explanation: Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.

Glossary of Terms

Packets Processed:	Total number of network packets analyzed by the NIDS during this session.
Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.
Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
port_scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a

target, which is often a precursor to an attack.

ddos_indicator:

Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.

unusual_protocol:

Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.

ml_anomaly:

The machine learning model detected traffic that is statistically unusual compared to normal patterns.